

# JAX-Based 2D Finite Element Solver: Volumetric Locking and Locking-Cure Techniques

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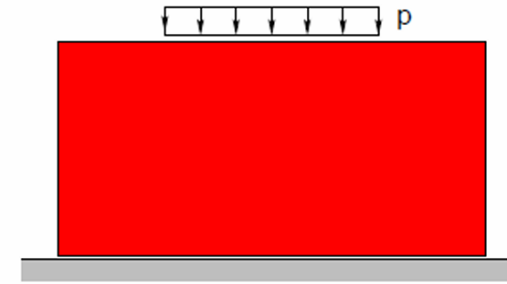
# The Challenge (Volumetric Locking)

**Incompressible Regime:** As  $\nu \rightarrow 0.4999$ , Lamé parameter  $\lambda \rightarrow \infty$ , acting as an extreme penalty constraint.

**Kinematic Over-Constraint:** Standard Q4 displacement elements use bilinear shape functions that cannot satisfy  $\text{tr}(\varepsilon) = 0$  under bending.

**Spurious Stiffness:** Artificial volumetric energy falsely dominates the total potential energy.

**Physical Failure:** Elements "lock," producing an unphysically rigid response and under-predicting displacements.



Geometry and boundary conditions of an almost incompressible block

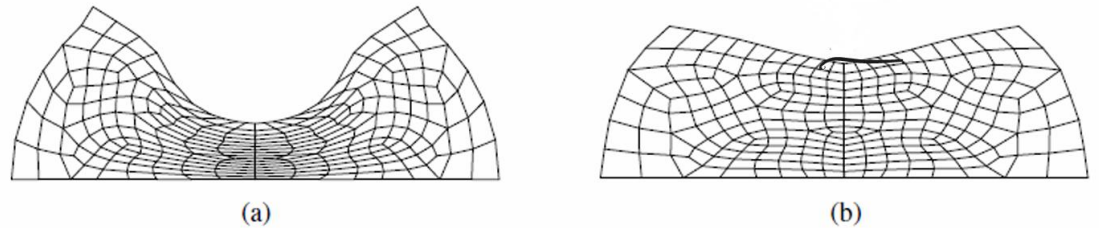


Figure 2.2: Deformed mesh for a given loading state, (a) result of a locking-free element formulation, (b) result of an element formulation which exhibits locking.

# 2D Block Indentation Benchmark

**Geometry:** 2D Rectangular Block ( $L = 10.0, H = 5.0$ )

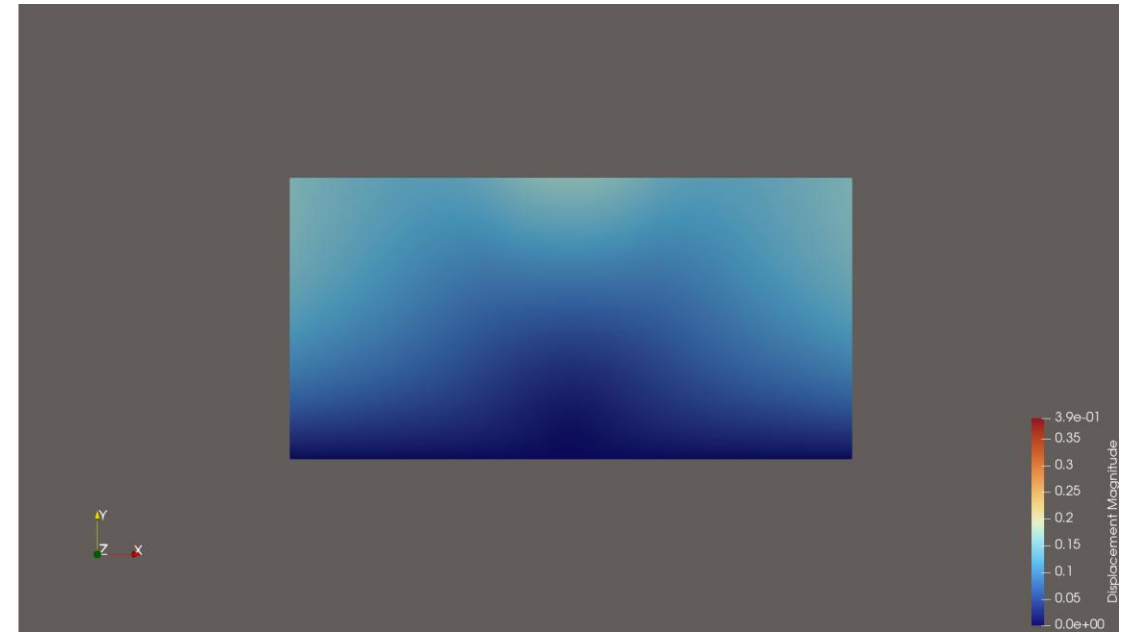
**Element Type:** Standard Bilinear Q4 Quadrilaterals

**Kinematics:** Plane Strain Assumption

**Material Constants:** Elastic Modulus  $E = 1000.0$ ,  
Critical Poisson's Ratio  $\nu = 0.4999$

**Displacement Boundary Conditions:** Fully Fixed  
Base ( $u_x = u_y = 0$  at  $Y = 0$ )

**Mechanical Loading:** Central Distributed Vertical  
Traction ( $q = -40.0$ , Span:  $3.75 \leq X \leq 6.25$ )



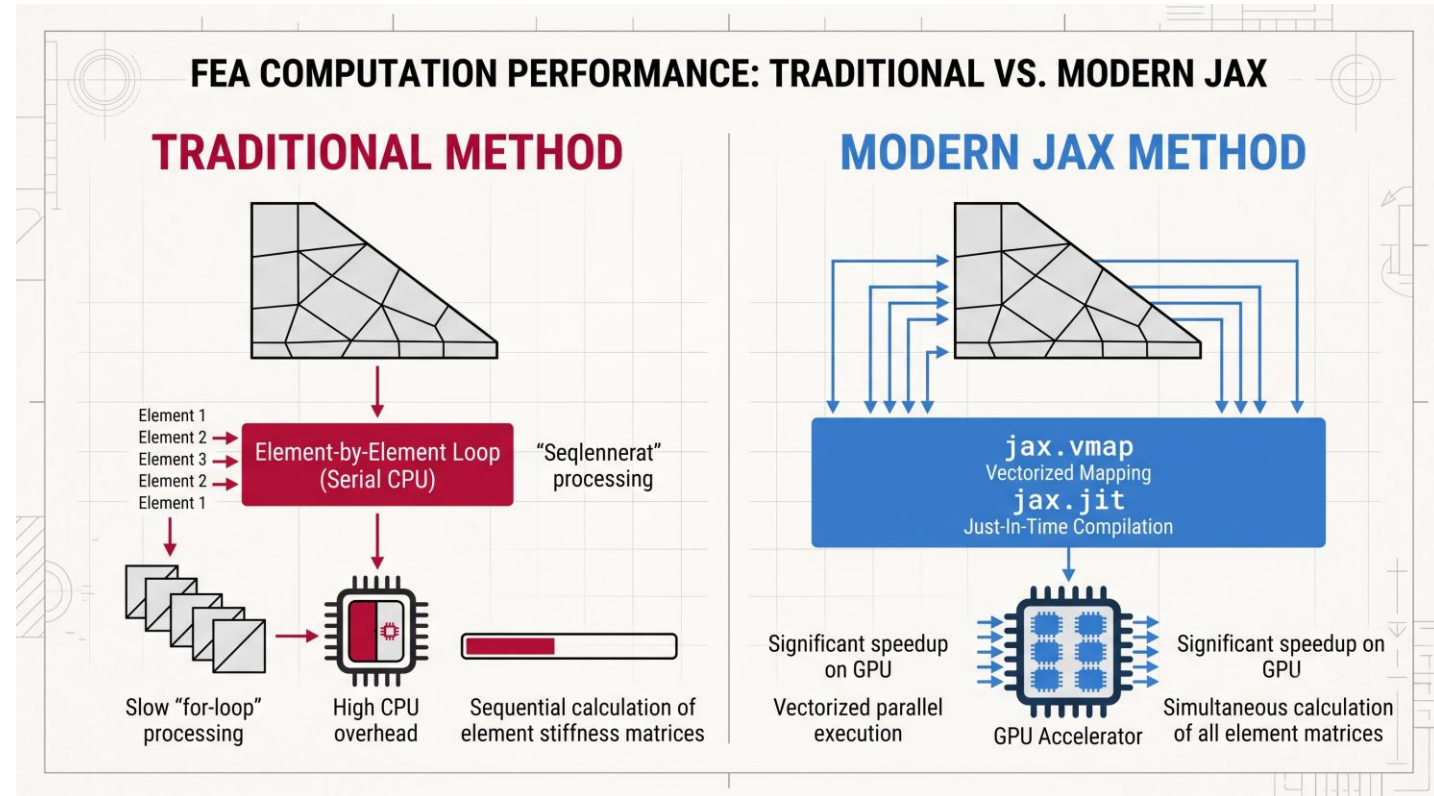
2D Block Indentation

# Computational Platform: Vectorized Element Mapping via JAX

**Loop Elimination:** Replaces traditional sequential element-by-element CPU loops to remove processing bottlenecks.

**Parallel Assembly (`jax.vmap`):** Automatically maps local kinematics and energy densities across all elements simultaneously.

**Exact Gradients (`jax.jacfwd`):** Physical strain tensors are computed via forward-mode Automatic Differentiation applied to shape functions.



# Locking Cures: SRI

## Selective Reduced Integration (SRI):

Decouples the strain energy density into deviatoric/shear and volumetric components.

Shear energy is integrated via a

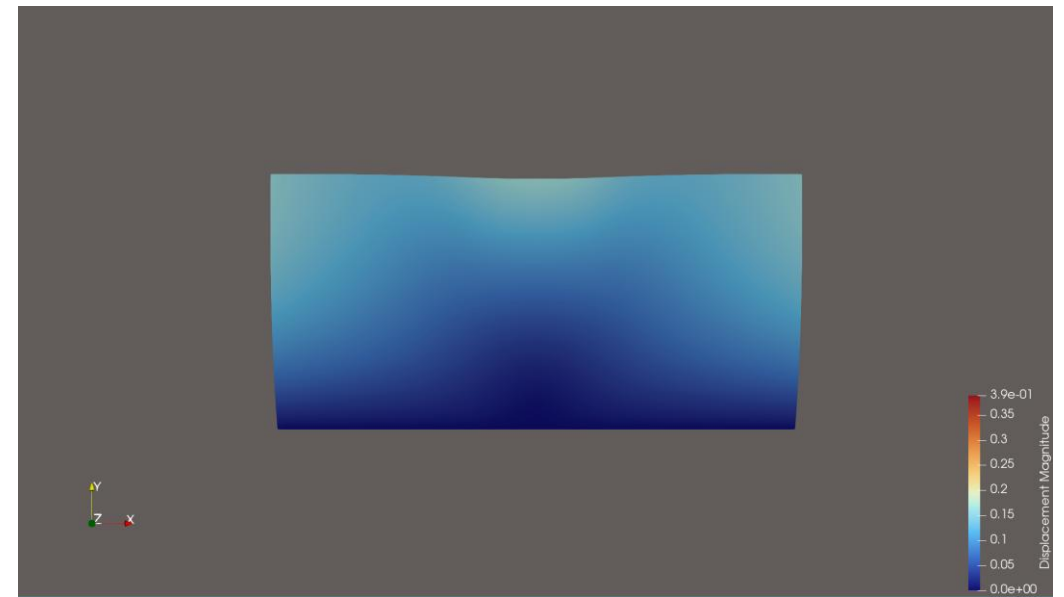
$$B = B_{dev} + B_{vol}$$

**Deviatoric Field  $B_{dev}$**  : Integrated using a standard **2×2 Gauss rule** to accurately capture bending and shear mechanics.

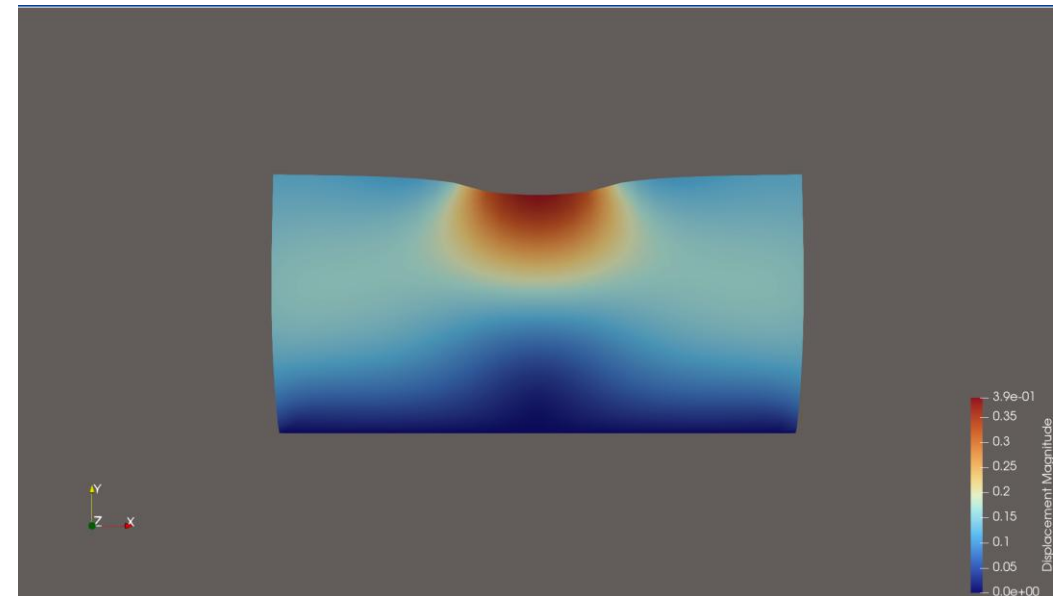
**Volumetric Field  $B_{vol}$**  : Under-integrated using a **Reduced 1×1 Gauss rule** evaluated strictly at the element centroid.

## Severe Sensitivity to Element

**Distortion:** In skewed or tapered geometries evaluating the Jacobian determinant  $\det J$  at a single isolated central point fails to capture spatial volume changes, inducing numerical noise and hourglass instabilities.



Displacement Field via Standard Q4



Displacement Field via SRI

## Algorithm 1: Element Stiffness Matrix (K<sub>e</sub> Formulation)

**Input** : Nodal coordinates X<sub>e</sub>, Material properties (E,  $\nu$ ), SRI Toggle

**Output**: Element stiffness matrix K<sub>e</sub>

1. **Initialize** K<sub>e</sub> matrix and standard material matrix E<sub>mat</sub>.
2. Integrate DEVIATORIC (shear) component using Full 2x2 Gauss rule.
3. **if** use\_sri == **False** then
4.     Integrate VOLUMETRIC component using Full 2x2 Gauss rule (Causes Locking)
5.     **else**
6.     Integrate VOLUMETRIC component using Reduced 1x1 rule at element center (Locking-Cure).
7.     **end if**
8.     Assemble combined components and return K<sub>e</sub>.

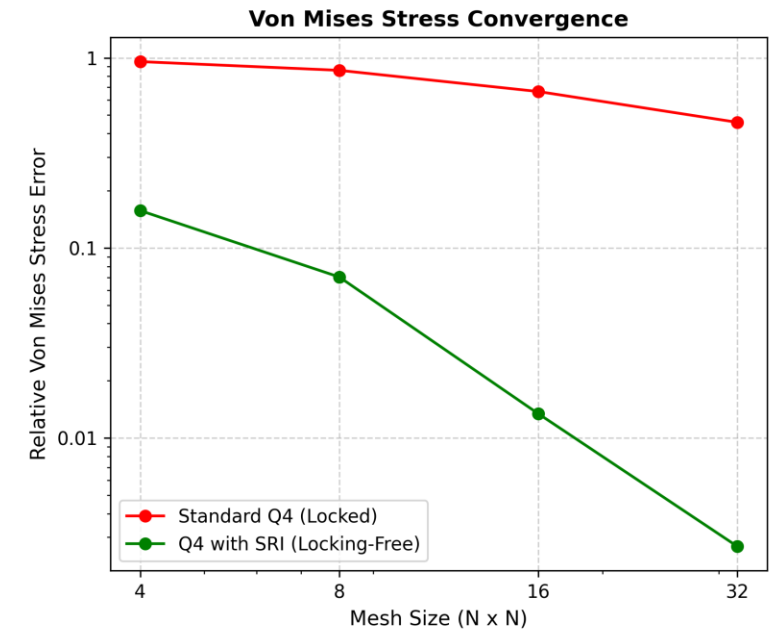
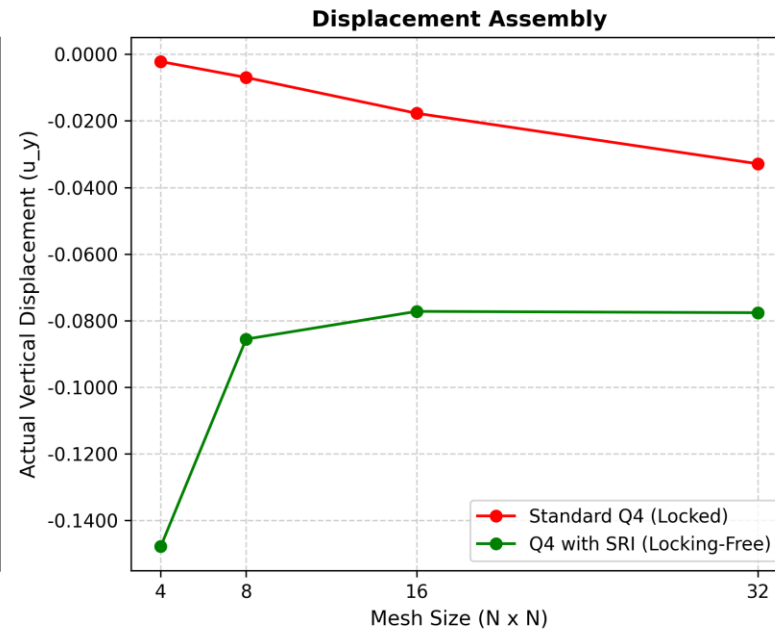
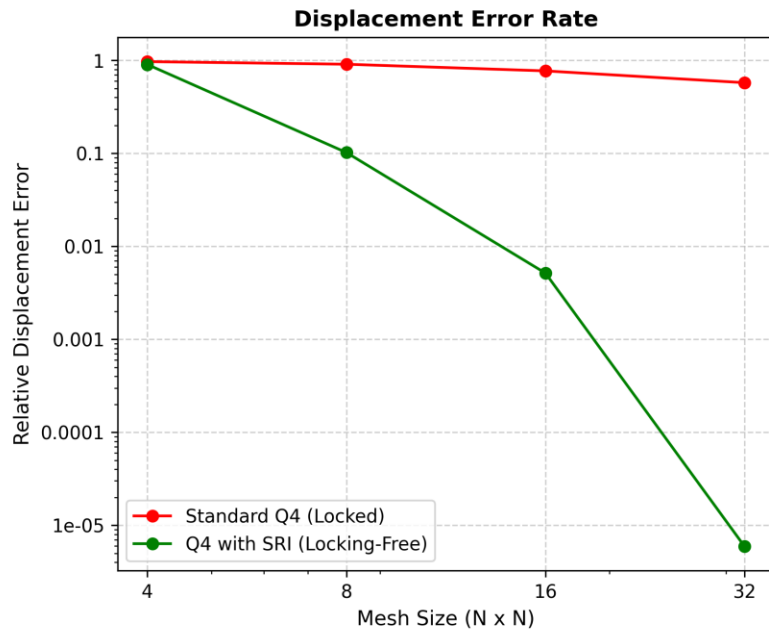
## Algorithm 2: JAX-Based FEM Convergence Pipeline

**Input** : Geometry (L, H), Loading (q), Mesh sizes N = [4, 8, 16, 32]

**Output**: Convergence charts & ParaView VTK files

1. Set physical constants and compute 64x64 high-res reference solution
2. **for each** mesh size N in [4, 8, 16, 32] **do**
3.     Generate N x N structured Q4 mesh and apply boundary conditions
4.     **for each** formulation in ["Locked", "SRI"] **do**
5.         Compute K<sub>elements</sub> in parallel using JAX vmap
6.         Assemble global matrix K and solve for displacements
7.         Calculate displacement & Von Mises stress errors against reference.
8.     **end for**
9.     **if** N == 32 then Export VTK files for ParaView visualization
10.    **end for**
11. Plot 3-panel convergence charts (Displacement Error, Value, and Stress Error).

# Convergence Results & Analysis



**Standard Q4 Failure:** Relative displacement and stress errors stagnate, confirming severe volumetric locking.

**SRI Convergence:** Relative errors drop sharply, and vertical displacement rapidly converges to the physical baseline ( $\approx -0.08$ ).

**Stress Recovery:** Standard Q4 suppresses stresses, whereas SRI captures smooth, high-fidelity Von Mises stress concentrations under the load patch.

# Future Work & Outlook

**Dimensional Scaling:** Extending the 2D plane strain architecture to 3D trilinear hexahedral (H8) continuum elements.

**Kinematic Extension:** Integrating the SRI formulation into large deformation regimes and hyper elastic laws (e.g., Neo-Hookean).

**Comparative Studies:** Benchmarking SRI against Enhanced Assumed Strain (EAS) and B bar formulations.

Thank you for your attention